Topic 6 Replica theory of spin glasses

- Let $\vec{s}^{\,0}$ be an N dimensional hidden spin configuration $s_i^0 \in \{\pm 1\}$ $i = 1 \ldots N$

- we don't get to measure $\vec{s}^{\,0}$; we only get to measure noisy correlations

$$J_{ij} = \underbrace{\frac{\lambda}{N} s_i^0 s_j^0}_{\text{signal}} + \underbrace{E_{ij}}_{\text{noise}}$$

$E_{ij} \sim \mathcal{N}(0, 1/N)$

$E_{ij} = E_{ji}$

signal to noise ratio (SNR)

$\boxed{SNR = \lambda}$

- how can we extract $\vec{s}^{\,0}$ from J ?

- Bayesian inference assuming uniform prior $p(\vec{s}) = \frac{1}{2^N}$:

$$P(\vec{s} \mid J) = \frac{p(J \mid \vec{s})\, p(\vec{s})}{P(J)} \qquad \text{Bayes rule}$$

$p(E_{ij}) \propto e^{-\frac{N}{2} E_{ij}^2}$

$$p(\vec{s} \mid J) \propto p(J \mid \vec{s}) = \prod_{ij} e^{-\frac{N}{2}\left(J_{ij} - \frac{\lambda}{N} s_i s_j\right)^2}$$

$$\propto e^{+\sum_{ij} \lambda J_{ij} s_i s_j - \frac{\lambda^2}{N} \sum_{ij} \cancelto{1}{(s_i s_j)^2}}$$

$$\propto e^{\lambda \sum_{ij} J_{ij} s_i s_j}$$

Now let $\boxed{E(\vec{s}) = -\sum_{1 \le i < j \le N} J_{ij} s_i s_j}$

$$\boxed{p_\beta(\vec{s}) = \frac{e^{-\beta E(\vec{s})}}{Z}}$$

Gibbs distribution $p_\beta(\vec{s})$ is posterior distribution $p(\vec{s} \mid J)$ iff $\beta = \lambda$

or $kT = \frac{1}{\lambda}$

temperature is inverse SNR

- high SNR → trust data J
  → freeze spin config $\vec{s}$ onto low energy states $E(\vec{s})$

- low SNR → opposite

Estimation

Let $\hat{s} = \langle \vec{s} \rangle_{p_\beta}$ be an estimator of s^0

How good is it? Could try to compute overlap

$$m \equiv \frac{1}{N} \hat{s} \cdot \vec{s}^{\,0}$$

as $N \to \infty$ m will concentrate to a deterministic quantity $m(\beta, \lambda)$

to determine it lets calculate free energy of $P_\beta(\vec{s})$

$$-\beta F(\beta,\lambda;E,\vec{s}^*) = \ln Z$$

$$= \ln \sum_{\vec{s}} e^{-\beta E(\vec{s})}$$

$$-\beta F(\beta,\lambda;E,\vec{s}^*) = \ln \sum_{\vec{s}} e^{+\beta \sum_{i<j} \left[\frac{\lambda}{N} s_i^* s_j^* + E_{ij}\right] s_i s_j}$$

now as $N\to\infty$ $-\beta F(\beta,\lambda;E,\vec{s}^*)$ will concentrate about its average:

$$-\beta F(\beta,\lambda) \equiv \langle -\beta F(\beta,\lambda;E,\vec{s}^*)\rangle_{E,\vec{s}^*}$$

- so we need to compute $\langle \ln Z\rangle$
- use the replica trick $Z^n = e^{n \ln Z}$

as $n\to 0$ $Z^n \approx 1 + n\ln Z$

so $\boxed{\ln Z = \lim_{n\to 0} \frac{Z^n - 1}{n}}$ Replica trick

$$\langle \ln Z\rangle = \frac{\langle Z^n\rangle - 1}{n}$$

$$Z = \int d\vec{s}\, e^{-\beta E(\vec{s})} \Rightarrow Z^n = \int \prod_{a=1}^{n} d\vec{s}^a\, e^{-\sum_{a=1}^{n} E(\vec{s}^a)}$$

$\vec{s} \to \vec{s}^1, \dots, \vec{s}^n$ n replicated decoupled systems

Goal: compute $\langle Z^n\rangle$ and take $n\to 0$ (keep terms linear in n in exponent)

let $\vec{s}^* = \vec{s}^0$

$$\langle Z^n\rangle_{s^0,E} = \sum_{s^0}\sum_{s^1}\cdots\sum_{s^n}\left\langle e^{\beta \sum_{i<j=1}^{N}\sum_{a=1}^{n}\left[\frac{\lambda}{N} s_i^0 s_j^0 + E_{ij}\right] s_i^a s_j^a}\right\rangle_{E_{ij}}$$

$$\frac{\lambda}{N}\sum_{i<j}^{N}\sum_{a=1}^{n} s_i^0 s_j^0 s_i^a s_j^a$$

$$= \frac{\lambda}{N}\sum_{a=1}^{n}\sum_{i<j}(s_i^0 s_i^a)(s_j^0 s_j^a)$$

$$\sum_{i<j} = \underbrace{\frac{1}{2}\sum_{i,j}}_{O(N^2)} - \underbrace{\sum_i}_{O(N)}$$

can drop

$$= \frac{\lambda}{2N}\sum_{a=1}^{n}\sum_{i,j}(s_i^0 s_i^a)(s_j^0 s_j^a)$$

$$= \frac{\lambda}{2N} \sum_{a=1}^{n} \left[\sum_i s_i^0 s_i^a\right]$$

$$= N\frac{\lambda}{2}\sum_{a=1}^{n} \left[\frac{1}{N} \sum_i s_i^0 s_i^a\right]^2$$

$$= N\frac{\lambda}{2}\sum_{a=1}^{n} (m^a)^2$$

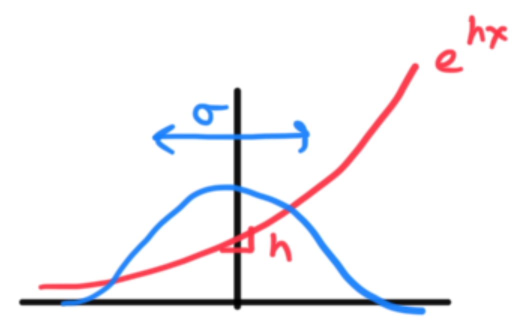
$$\boxed{m^a \equiv \frac{1}{N}\sum_{i=1}^{N} s_i^0 s_i^a}$$

$$\boxed{\langle Z^n\rangle_{s_0,E} = \sum_{\{s_i^a\}} \underbrace{e^{N\beta\frac{\lambda}{2}\sum_{a=1}^{n} (m^a)^2}}_{\text{signal term}} \underbrace{\left\langle e^{\beta\sum_{a=1}^{n}\sum_{i,j=1}^{N} E_{ij}\, s_i^a s_j^a}\right\rangle_{E_{ij}}}_{\text{noise term}}}$$

tries to pull each replica s^a towards s^0

need to avg over gaussian E_{ij} : use Gaussian integral identity

$$\boxed{\langle e^{hx}\rangle_x = e^{\sigma^2\frac{h^2}{2}} \quad \text{when } x\sim \mathcal{N}(0,\sigma^2)}$$



$$\left.\begin{array}{l} x\sim E_{ij} \\ \sigma^2\sim 1/N \\ h\sim \beta\sum_{a=1}^{n} s_i^a s_j^a \end{array}\right\} \Rightarrow \left\langle e^{\beta\left[\sum_{a=1}^{n} s_i^a s_j^a\right] E_{ij}}\right\rangle_{E_{ij}} = e^{\frac{\beta^2}{2N}\left[\sum_{a=1}^{n} s_i^a s_j^a\right]^2}$$

e^{σ^2}

so

$$\left\langle e^{\beta\sum_{a=1}^{n}\sum_{i<j}^{N} E_{ij}\, s_i^a s_j^a}\right\rangle = e^{\frac{\beta^2}{2N}\sum_{i<j}\left[\sum_{a=1}^{n} s_i^a s_j^a\right]^2}$$

now

$$\sum_{i<j}\left[\sum_{a=1}^{n} s_i^a s_j^a\right]^2$$

$$= \sum_{i<j}\sum_{a,b=1}^{n} s_i^a s_j^a s_i^b s_j^b$$

$$= \sum_{a,b=1}^{n}\sum_{i<j} s_i^a s_i^b s_j^a s_j^b$$

$$= \sum_{a,b=1}^{n}\frac{1}{2}\sum_{i,j} s_i^a s_i^b s_j^a s_j^b - \underbrace{\sum_{a,b=1}^{n}}_{n^2}\underbrace{\sum_{i=1}^{N}}_{N}\underbrace{s_i^a s_i^b s_i^a s_i^b}_{1}$$

ignore $O(n^2 N)$

$$\approx \frac{1}{2}\sum_{a,b}\left(\sum_i s_i^a s_i^b\right)^2$$

$$\boxed{Q_{ab} = \frac{1}{N}\sum_{i=1} s_i^{a} s_i^{b}}$$

$$= \frac{N^2}{2} \sum_{a,b} Q_{ab}^2$$

$$= N^2 \sum_{a<b} Q_{ab}^2 + \frac{N^2}{2}\sum_a Q_{aa}$$

$$= N^2 \sum_{a<b} Q_{ab}^2 + \frac{N^2 n}{2}$$

so

$$\boxed{\left\langle e^{\beta \sum_{a=1}^{n} \sum_{i<j}^{N} E_{ij} s_i^a s_j^a} \right\rangle_E = e^{\frac{\beta^2 N n}{4} + \frac{\beta^2 N}{2} \sum_{a<b} Q_{ab}^2}}$$

so

$$\boxed{\langle Z^n \rangle_{s_0, E} = \sum_{\{s_i^a\}} e^{\underbrace{N \frac{\beta\lambda}{2} \sum_{a=1}^{n} (m^a)^2}_{\text{signal term}} + \underbrace{N \frac{\beta^2 n}{4} + N \frac{\beta^2}{2} \sum_{a<b} Q_{ab}^2}_{\text{noise term}}}}$$

signal term: pull s^a towards s^0

$$\boxed{m^a = \frac{1}{N} \sum_{i=1}^{N} s_i^0 s_i^a}$$

noise term: pull s^a towards each other

$$\boxed{Q_{ab} = \frac{1}{N} \sum_{i=1}^{N} s_i^a s_i^b}$$

<u>Note:</u> conditioned on disorder E replicas are independent:

$$P(\vec{s}^1, \ldots, \vec{s}^n \mid E, \vec{s}^0) = \prod_{a=1}^{n} P(s^a \mid E, s^0)$$

but marginalizing out disorder, replicas are coupled

$$\underbrace{P(\vec{s}^1, \ldots, \vec{s}^n \mid s^0)}_{} = \int dE\, P(E)\, P(\vec{s}^1, \ldots, \vec{s}^n \mid E, s^0)$$

Marginal over replicas does <u>NOT</u> factorize

replicas are <u>coupled</u> after marginalizing out noise...

now the sum in $\langle Z^n \rangle$ depends on $\vec{s}^0, \vec{s}^1, \ldots, \vec{s}^n$ only through the $(n+1) \times (n+1)$ matrix of inner products Q'_{ab} $a,b = 0 \ldots n$

$$Q' = \begin{bmatrix} 1 & m^1 & \cdots\cdots & m^n \\ m^1 & 1 & & \\ \vdots & & 1 & Q_{ab} \\ \vdots & & 1 & \\ m^n & Q_{ab} & & 1 \end{bmatrix} \qquad Q'_{ab} = \begin{cases} 1 & a=b \\ m^b & a=0 \quad b>0 \\ Q_{ab} & a\neq b \quad a,b>0 \end{cases}$$

so lets sum over $s^0, \ldots s^n$ by:

1) first summing over all $s^0, \ldots s^n$ with a given Q'_{ab}

2) <u>then</u> integrating over Q'_{ab}

the way to do this is introducing δ-functions

$$1 = \int dQ'_{ab}\, \delta\left[\sum_{i=1}^{N} s_i^a s_i^b - NQ'_{ab}\right] \qquad 0 \leq a < b \leq n$$

let $$\boxed{E(Q') \equiv -\left(\frac{\beta\lambda}{2}\sum_{a=1}^{n} m_a^2 + \frac{\beta^2 n}{4} + \frac{\beta^2}{2}\sum_{1\leq a<b\leq n} Q_{ab}^2\right)}$$

<u>then</u> $$\langle Z^n \rangle = \sum_{s_i^0} \cdots \sum_{s_i^n} e^{-NE(Q')}$$

$$= \int \prod_{0\leq a<b\leq n} dQ'_{ab}\, e^{-NE(Q')} \underbrace{\sum_{\vec{s}^0} \cdots \sum_{\vec{s}^n} \delta\left[\sum_{i=1}^{N} s_i^a s_i^b - NQ'_{ab}\right]}$$

how to compute volume of spin configs with a given overlap?

introduce exponential rep of δ-function

<u>recall</u>: $$\boxed{\delta(x) = \int \frac{d\hat{x}}{2\pi} e^{i\hat{x}x}\, dx}$$

$$\boxed{\delta\left[\sum_{i=1}^{N} s_i^a s_i^b - NQ'_{ab}\right] = \int \frac{d\hat{Q}'_{ab}}{2\pi} e^{\hat{Q}'_{ab}\left[\sum_{i=1}^{N} s_i^a s_i^b - NQ'_{ab}\right]}}$$

so let

$$\boxed{\sum_{s^0} \cdots \sum_{s^n} \prod_{0\leq a<b\leq n} \delta\left[\sum_{i=1}^{N} s_i^a s_i^b - NQ'_{ab}\right] \equiv e^{NS[Q']}}$$

assume Large deviation principle

$S[Q']$ = entropy of spin

configs $s^0, \dots s^n$

with given overlap Matrix Q'

now lets calculate $S[Q]$:

$$e^{NS[Q']} = \sum_{\vec{s}^0} \cdots \sum_{\vec{s}^n} \prod_{0\le a<b\le n} \delta\left[\sum_{i=1}^{N} s_i^a s_i^b - N Q'_{ab}\right]$$

$$= \sum_{\vec{s}^0} \cdots \sum_{\vec{s}^n} \int \prod_{0\le a\le b\le n} d\hat{Q}_{ab}\, e^{\sum_{0\le a<b\le n} -N \hat{Q}'_{ab} Q'_{ab} + \hat{Q}'_{ab} \sum_{i=1}^{N} s_i^a s_i^b}$$

$$= \int \prod_{0\le a<b\le n} d\hat{Q}_{ab}\, e^{-N \sum_{0\le a<b\le n} \hat{Q}'_{ab} Q'_{ab}} \underbrace{\sum_{\vec{s}^0} \cdots \sum_{\vec{s}^n} e^{\sum_{0\le a<b\le n} \hat{Q}'_{ab} \sum_{i=1}^{N} s_i^a s_i^b}}_{\|}$$

$s^0, \dots s^n$ are N dimensional vectors

big difference

$$\sum_{\vec{s}^0} \cdots \sum_{\vec{s}^n} \prod_{i=1}^{N} e^{\sum_{0\le a<b\le n} \hat{Q}'_{ab} s_i^a s_i^b}$$

$$= \prod_{i=1}^{N} \left[\sum_{s_i^0} \cdots \sum_{s_i^n} e^{\sum_{0\le a<b\le n} \hat{Q}'_{ab} s_i^a s_i^b}\right]$$

sites are decoupled!

now $s^0, \dots s^n$ are scalars ± 1

$$= \left[\sum_{s^0} \cdots \sum_{s^n} e^{\sum_{0\le a<b\le n} \hat{Q}'_{ab} s^a s^b}\right]^N$$

$$\equiv Z[\hat{Q}']^N$$

$$= e^{N \ln Z[\hat{Q}']}$$

$$\equiv e^{N \Phi[\hat{Q}']}$$

$$\boxed{\underbrace{\Phi[\hat{Q}']}_{\text{(neg) free energy}} = \ln \underbrace{Z[\hat{Q}']}_{\text{partition function of a replica system}} = \ln \sum_{s^0 \dots s^n} e^{\sum_{0\le a<b\le n} \hat{Q}'_{ab} s^a s^b}}$$

so....

$$e^{NS(Q')} = \int \prod_{0\le a<b\le n} d\hat{Q}'_{ab}\, e^{N\left\{-\sum_{0\le a<b\le n} \hat{Q}'_{ab} Q'_{ab} + \Phi[\hat{Q}']\right\}}$$

saddle point approx to integral over $\hat{Q}'_{ab}$:

$$\boxed{S[Q'] = \max \left\{-\sum_{0\le a<b\le n} \hat{Q}'_{ab} Q'_{ab} + \Phi[\hat{Q}']\right\}}$$

$$\underset{Q'}{}$$

Entropy $S[Q']$ is Legendre dual of free energy $\Phi[\hat{Q}']$

extremum condition : $\frac{\partial}{\partial \hat{Q}'_{ab}}$:

$$\boxed{Q'_{ab} = \frac{\partial \Phi}{\partial \hat{Q}'_{ab}} = \langle s^a s^b \rangle_{\hat{Q}'}}$$

note : $$\frac{\partial \Phi}{\partial \hat{Q}'_{ab}} = \frac{\partial}{\partial \hat{Q}'_{ab}} \ln Z[\hat{Q}'_{ab}]$$

$$= \sum_{s^0, \dots s^n} \frac{1}{Z[\hat{Q}'_{ab}]} s^a s^b e^{\sum_{0 \le a < b \le n} \hat{Q}'_{ab} s^a s^b}$$

$$\equiv \langle s^a s^b \rangle_{\hat{Q}_{ab}}$$

where $\langle \cdot \rangle_{\hat{Q}_{ab}}$ is an average w.r.t. to a boltzmann distribution over

$n+1$ spins $s^0 \dots s^n$ with energy function

$$\boxed{\begin{aligned} E(s^0, \dots, s^n) &= -\sum_{0 \le a < b \le n} \hat{Q}_{ab} s^a s^b \\ p(s^0, \dots s^n) &= \frac{e^{-E(s^0, \dots s^n)}}{Z[\hat{Q}'_{ab}]} \end{aligned}}$$

Now lets get back to spin glass replica problem

recall:

$$\langle z^n \rangle = \int \prod_{0 \le a < b \le n} dQ'_{ab} \, e^{-N E[Q']} e^{N S[Q']}$$

$$\boxed{\langle z^n \rangle = \int \prod_{0 \le a < b \le n} dQ'_{ab} \, d\hat{Q}'_{ab} \, e^{N \left\{ -E[Q'] \underbrace{- \sum_{0 \le a < b \le n} \hat{Q}'_{ab} Q'_{ab} + \Phi[\hat{Q}'] }_{= + S[Q'] \text{ after extremizing over } \hat{Q}'} \right\}}}$$

at large N can evaluate via a saddle point approx

extremum conditions :

Correlations $\leftarrow$ Coupling constant

$\frac{\partial}{\partial \hat{Q}'_{ab}}$: $Q'_{ab} = \langle s^a s^b \rangle_{\hat{Q}'_{ab}}$

$\frac{\partial}{}$: $\hat{Q}'_{ab} = \frac{\partial E}{}$

$$\frac{\partial}{\partial Q'_{ab}} \quad : \qquad \boxed{\qquad \hat{Q}_{ab} \qquad\qquad \partial Q'_{ab}}$$

coupling constant $\leftarrow$ correlations

$\hat{Q}'_{ab}$ are coupling constants

Q'_{ab} are correlations

Physical Meaning of order parameters Q'_{ab}

- what does order parameter Q_{ab} mean ??
- consider two iid samples $\vec{s}^1$ and $\vec{s}^2$ from $P_\beta(\vec{s}) \propto e^{-\beta E(\vec{s})}$
- let $q = \frac{1}{N}\sum_{i=1}^{N} s_i^1 s_i^2$ be there overlap
- what is the distribution of q :
- $$P(q \mid E, \vec{s}^0) = \frac{1}{Z(\beta,\lambda,E,s^0)^2} \sum_{\vec{s}^1}\sum_{\vec{s}^2} \delta\left[q - \frac{1}{N}\sum_{i=1}^{N} s_i^1 s_i^2\right] e^{-\beta E(s^1) - \beta E(s^2)}$$
- we would like to compute
  $$P(q) \equiv \left\langle P(q \mid E, \vec{s}^0)\right\rangle_{E,\vec{s}^0}$$
- but it is hard to average $\frac{1}{Z^2}$
- so we under replica trick : $Z^{-2} = \lim_{n\to 0} Z^{n-2}$
- so introduce $n-2$ <u>additional</u> replicas plus <u>real</u> replicas $\vec{s}^1$ and $\vec{s}^2$ (and $\vec{s}^0$)
- then

$$P(q) = \lim_{n\to 0}\left\langle \sum_{s^0}\cdots\sum_{s^n} \delta\left[q - \frac{1}{N}\sum_{i=1}^{N} s_i^1 s_i^2\right] e^{-\beta\sum_{a=1}^{n} E(\vec{s}^a)}\right\rangle_{E,s^0}$$

- after a (long) sequence of steps similar to above we find

$$\boxed{P(q) = \lim_{n\to 0} \frac{1}{n(n-1)} \sum_{1\le a\neq b\le n} \delta[q - Q_{ab}]}$$

where Q_{ab} obeys saddle point equations above

- distribution of overlaps, averaged over s^0, and E, = fraction of elements of Q_{ab} taking a given value

- Similarly, pick a single spin configuration $\vec{s}^1 \sim P_\beta(\vec{s})$
  - let $M \equiv \frac{1}{N}\sum_{i=1}^{N} s_i^0 s_i^1$

- the distribution of m, averaged over E and $\vec{S}^0$ is:

where m^a satisfies saddle point equations above

$$\boxed{P(m) = \lim_{n \to 0} \frac{1}{n} \sum_{a=1}^{n} \delta\left[m - m^a\right]}$$

- distribution of overlap with $\vec{S}^0$, averaged over E and $\vec{S}^0$
  = fraction of elements of m^a taking a particular value

next time: take $n \to 0$ limit, solve saddle point equations, assuming replica symmetry